Knowledge Sharing Portal (Phase -1)

Tech Stack :

Frontend:

- React JS
- React Router
- Redux / Context API
- Tailwind CSS / Material UI
- Axios
- React Hook Form
- Vite / Create React App

Backend:

- Node.js
- Express.js
- REST APIs
- JWT Authentication

Database:

- MongoDB

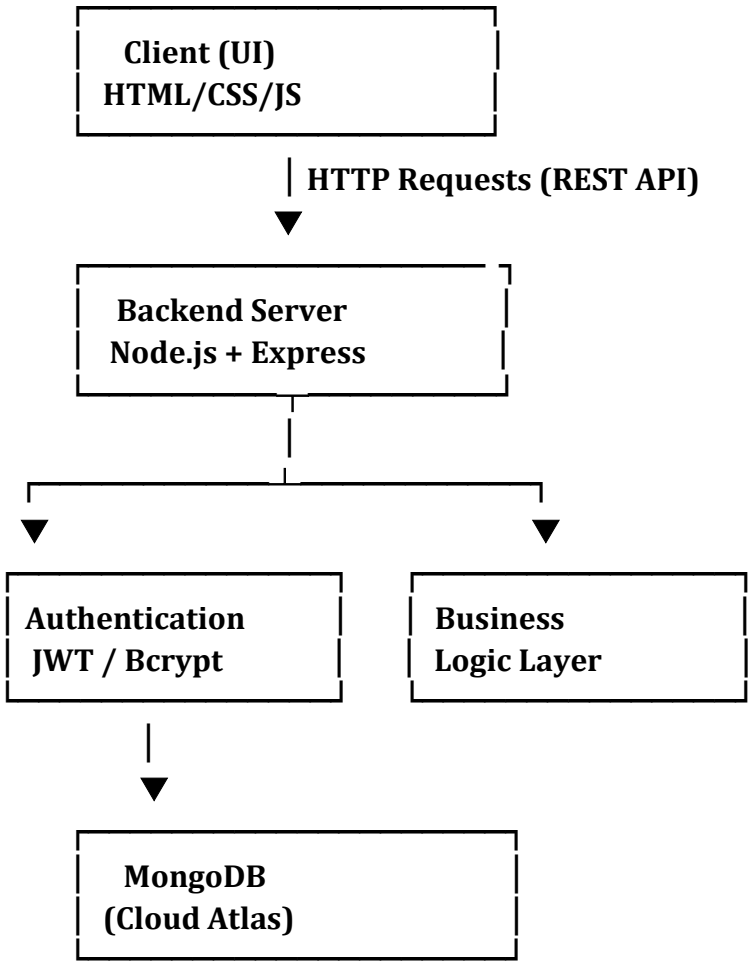
DevOps & Deployment:

- Git & GitHub
- Vercel / Netlify
- MongoDB Atlas

Security:

- Password Hashing (bcrypt)
- Role-Based Authorization
- Input Validation
- API Rate Limiting

Architecture:



Database Schema:

1) User Collection

```
User {
  _id: ObjectId,
  username: String,
  email: String,
  password: String,
  role: String, // user | moderator | admin
  skills: [String],
  points: Number,
  badges: [String],
  followers: [ObjectId],   // ref User
  following: [ObjectId],   // ref User
  enrolledPaths: [ObjectId], // ref LearningPath
  bookmarks: [ObjectId],   // ref Post
  createdAt: Date
}
```

2) Post Collection

```
Post {
  _id: ObjectId,
  title: String,
  content: String,
  type: String, // article | question | tutorial | discussion
  author: ObjectId, // ref User
  category: ObjectId, // ref Category
  tags: [ObjectId], // ref Tag
  views: Number,
  likes: Number,
  visibility: String, // public | private | followers
  createdAt: Date
}
```

3) Comment Collection

```
Comment {
  _id: ObjectId,
  content: String,
  author: ObjectId, // ref User
  post: ObjectId, // ref Post
  likes: Number,
  createdAt: Date
}
```

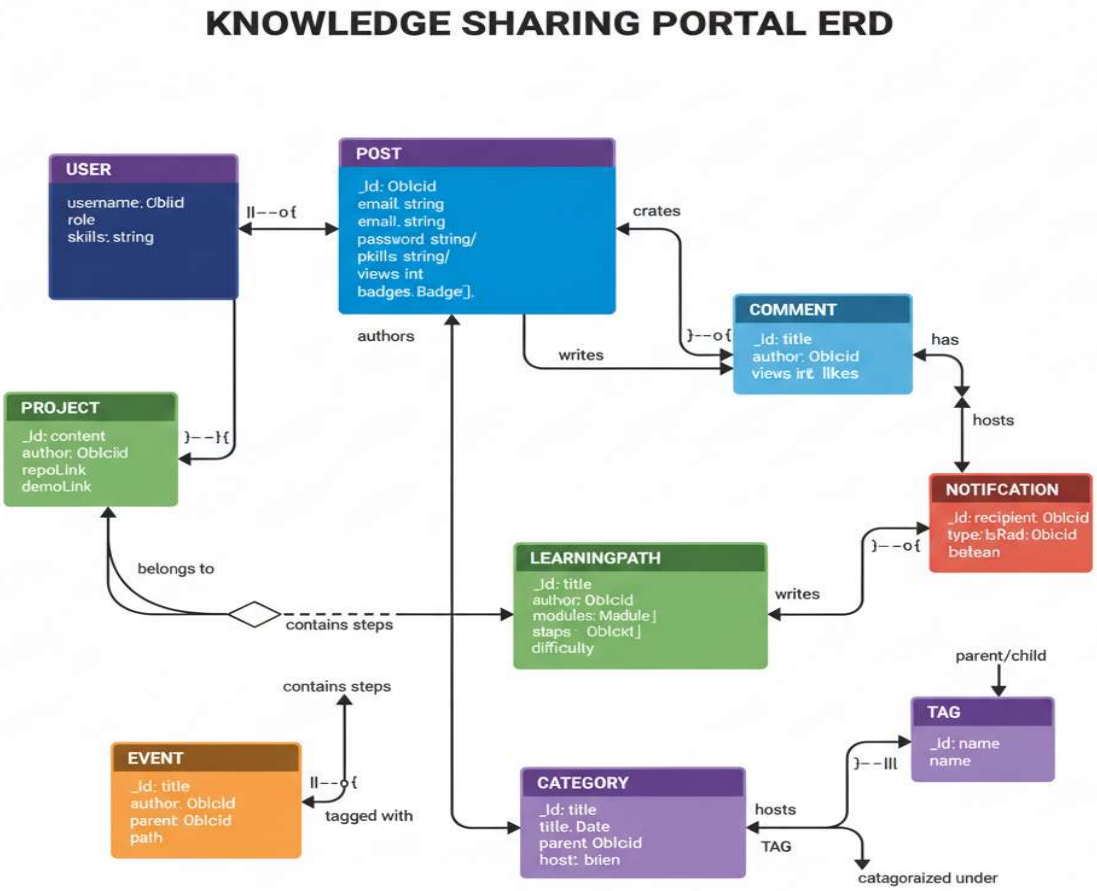
4) Project Collection

```
Project {
  _id: ObjectId,
  title: String,
  description: String,
  repoLink: String,
  demoLink: String,
  author: ObjectId, // ref User
  likes: [ObjectId], // ref User
  createdAt: Date
}
```

5) LearningPath Collection

```
LearningPath {
  _id: ObjectId,
  title: String,
  description: String,
  author: ObjectId, // ref User
  steps: [ObjectId], // ref Post
  difficulty: String, // Beginner | Intermediate | Advanced
  enrolledUsers: [ObjectId], // ref User
  createdAt: Date
}
```

Entity Diagram(ER):



Wire Frames:

A. Landing / Home Page Wireframe

```
| Logo | Home | Categories | About | Login |
-----------------------------------------------------
|              Hero Section                          |
|   "Share Knowledge. Grow Together."              ||
 [ Explore Posts ]   [ Create Post ]           |
-----------------------------------------------------
| Sidebar      |      Post Cards Grid               |
| - Tech       |   --------------------------       |
| - Cloud      |   | Post Card | Post Card |       ||
- AI         |   | Post Card | Post Card |       |
-------------------------------------------------------|
Footer                                               |
-----------------------------------------------------
```

B) Dashboard Page Wireframe

```
| Navbar (Search | Profile | Notifications)      |
-------------------------------------------------
| Sidebar       | Main Content                   |
| - Dashboard   |  Welcome, User                |
| - My Posts    |  Stats Cards (Total Posts)    |
| - Create Post |  Recent Posts List            ||
- Logout      |                              |
-------------------------------------------------
```

C. Post Detail Page Wireframe

```
| Navbar                                          |
--------------------------------------------------
| Post Title
|| Author | Date | Category                       |
| Post Content (Long Text Area)                  |
--------------------------------------------------
Like | Share | Comment                          |
--------------------------------------------------
| Comments Section                               |
|  - Comment 1
||  - Comment 2                                  |
--------------------------------------------------
```